

# Joint PDF factorization

From a generative model to local factors.

Point and extended objects.

Known and unknown object counts.

**GrBP:** Precluster, then associate whole groups.

**S1**

Point · known

**S2**

Point · unknown

**S3**

Extended · known

**S4**

Extended · unknown

# Two modeling choices shape the graph.

**Object model:** one measurement or several per scan.

**Object count:** fixed, or represented by potential objects and existence variables.

|          | Known count                | Unknown count                       |
|----------|----------------------------|-------------------------------------|
| Point    | <u>S1 · exclusivity</u>    | <u>S2 · existence + new objects</u> |
| Extended | <u>S3 · Poisson counts</u> | <u>S4 · anchored new objects</u>    |

GrBP · Preclustering → group association  
Appendix I of the EO manuscript

# Prior $\times$ allocation $\times$ measurements

1. Read the counts from the association.
2. Insert the Poisson count laws.
3. Cancel allocation factorials.
4. Collect constants; marginalize auxiliaries.

$$f(S, A, Z) = f(S) p(A, m \mid S) f(Z \mid A, m, S)$$

$S$  collects object states;  $A$  collects associations and any existence variables needed for the allocation.

# What is observed, and what is inferred?

**Observed:** the ordered measurement vector  $Z$  and its count.

**Inferred:** states, associations, and existence when applicable.

Clutter parameters and measurement models are supplied.

point:  $\mathbf{x}$       extended:  $\mathbf{y} = (\mathbf{x}, \mathbf{e})$

uncertain existence:  $r \in \{0, 1\}$

$$C(Z, m) = \frac{\mu_c^m e^{-\mu_c}}{m!} \prod_{j=1}^m f_c(\mathbf{z}_j)$$

For fixed data and clutter model,  $C(Z, m)$  is constant in the inferred states and assignments.

# Each target makes at most one claim.

$n_t$  existing point targets;  $m$  measurements.

A nonzero claim must be unique.  
Several targets may have the null claim.

$$a_i \in \{0, 1, \dots, m\}, \quad a_i = 0 : \text{missed detection}$$

$$\psi(\mathbf{a}) = \begin{cases} 1, & a_i \neq a_j \text{ whenever } i \neq j, a_i, a_j \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathcal{D} = \{i : a_i \neq 0\}, \quad n_c = m - |\mathcal{D}|$$

# Divide by the clutter explanation.

Claimed measurements use the target likelihood.

Unclaimed measurements use  $f_c$ .

Multiplying and dividing by clutter densities leaves a ratio for each detection.

$$\begin{aligned} f(Z \mid \mathbf{a}, \underline{\mathbf{X}}, m) &= \prod_{i \in \mathcal{D}} f(\mathbf{z}_{a_i} \mid \mathbf{x}_i) \prod_{j \in \mathcal{C}} f_c(\mathbf{z}_j) \\ &= \left[ \prod_{i \in \mathcal{D}} \frac{f(\mathbf{z}_{a_i} \mid \mathbf{x}_i)}{f_c(\mathbf{z}_{a_i})} \right] \prod_{j=1}^m f_c(\mathbf{z}_j). \end{aligned}$$

# A permutation removes the clutter factorial.

Given the detected targets, all consistent assignments to distinct measurement slots are uniform.

The clutter Poisson factorial cancels the allocation factorial.

$${}_m P_{m-n_c} = \frac{m!}{n_c!},$$

$$\frac{\mu_c^{n_c} e^{-\mu_c}}{n_c!} \frac{n_c!}{m!} = \frac{\mu_c^m e^{-\mu_c}}{m!} \prod_{i \in \mathcal{D}} \frac{1}{\mu_c}.$$

One factor  $1/\mu_c$  remains for each detected target.

# Detection ratios and missed detections

Each target contributes its prior and one local factor  $g$ .

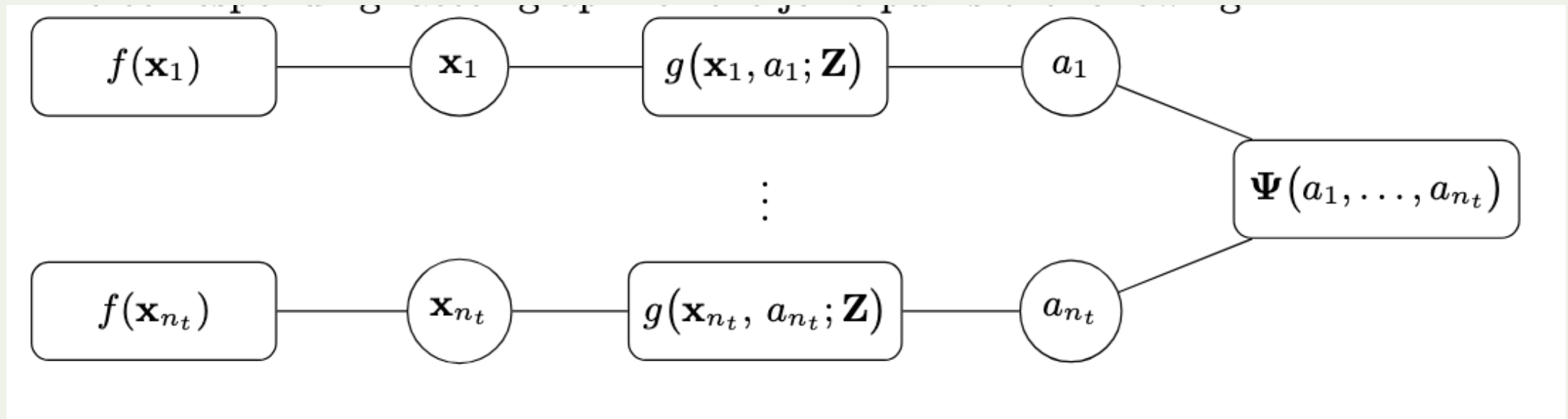
The consistency factor couples their association choices.

$$g(\mathbf{x}_i, a_i; Z) = \begin{cases} \frac{P_d(\mathbf{x}_i) f(\mathbf{z}_{a_i} | \mathbf{x}_i)}{\mu_c f_c(\mathbf{z}_{a_i})}, & a_i \neq 0, \\ 1 - P_d(\mathbf{x}_i), & a_i = 0. \end{cases}$$

$$f(\underline{\mathbf{X}}, \mathbf{a} | Z) \propto \psi(\mathbf{a}) \prod_{i=1}^{n_t} f(\mathbf{x}_i) g(\mathbf{x}_i, a_i; Z)$$

# One chain per target; one shared constraint.

Prior  $\rightarrow$  state  $\rightarrow$  local likelihood  $\rightarrow$  association.  $\Psi$  forbids duplicate nonzero claims.



Original Scenario 1 factor graph from the restored article.

# Augment each state with existence.

Legacy potential objects carry

$$\underline{\mathbf{y}}_i = (\underline{\mathbf{x}}_i, \underline{\mathbf{r}}_i).$$

An existing object can be detected or missed. A nonexistent object cannot claim a measurement.

$$\underline{\mathbf{g}}(\underline{\mathbf{x}}_i, \underline{\mathbf{r}}_i, a_i; Z) = \begin{cases} \frac{P_d(\underline{\mathbf{x}}_i) f(\mathbf{z}_{a_i} | \underline{\mathbf{x}}_i)}{\mu_c f_c(\mathbf{z}_{a_i})}, & a_i \neq 0, \underline{\mathbf{r}}_i = 1, \\ 1 - P_d(\underline{\mathbf{x}}_i), & a_i = 0, \underline{\mathbf{r}}_i = 1, \\ 0, & a_i \neq 0, \underline{\mathbf{r}}_i = 0, \\ 1, & a_i = 0, \underline{\mathbf{r}}_i = 0. \end{cases}$$

# Split unclaimed points into new objects and clutter.

Each measurement has a new potential object  $\bar{\mathbf{y}}_j$ .

On the valid support, new-object and clutter counts follow directly from existence and association.

$$\mathcal{N} = \{j : \bar{r}_j = 1\}, \quad n_n = |\mathcal{N}|, \quad n_c = m - |\mathcal{D}| - n_n$$

$$\binom{n_c + n_n}{n_n}^{-1} ({}_m P_{m-n_c-n_n})^{-1} = \frac{n_n! n_c!}{m!},$$

$$\mu_n^{n_n} \mu_c^{n_c} = \mu_c^m \left( \frac{\mu_n}{\mu_c} \right)^{n_n} \prod_{i \in \mathcal{D}} \frac{1}{\mu_c}.$$

# A claimed point cannot also start a new object.

$\mathcal{T}$  contains points already claimed by legacy objects.

A valid new-object branch needs both the birth intensity  $\mu_n$  and state density  $f_n$ .

$$\bar{g}(\bar{\mathbf{x}}_j, \bar{r}_j, \mathbf{a}; \mathbf{z}_j) = \begin{cases} \frac{\mu_n f_n(\bar{\mathbf{x}}_j) f(\mathbf{z}_j | \bar{\mathbf{x}}_j)}{\mu_c f_c(\mathbf{z}_j)}, & j \notin \mathcal{T}, \bar{r}_j = 1, \\ 0, & j \in \mathcal{T}, \bar{r}_j = 1, \\ f_d(\bar{\mathbf{x}}_j), & \bar{r}_j = 0. \end{cases}$$

# Legacy chains plus new-object factors

The same  $\psi(\mathbf{a})$  enforces unique legacy claims.

Each new-object factor depends on the entire legacy association vector.

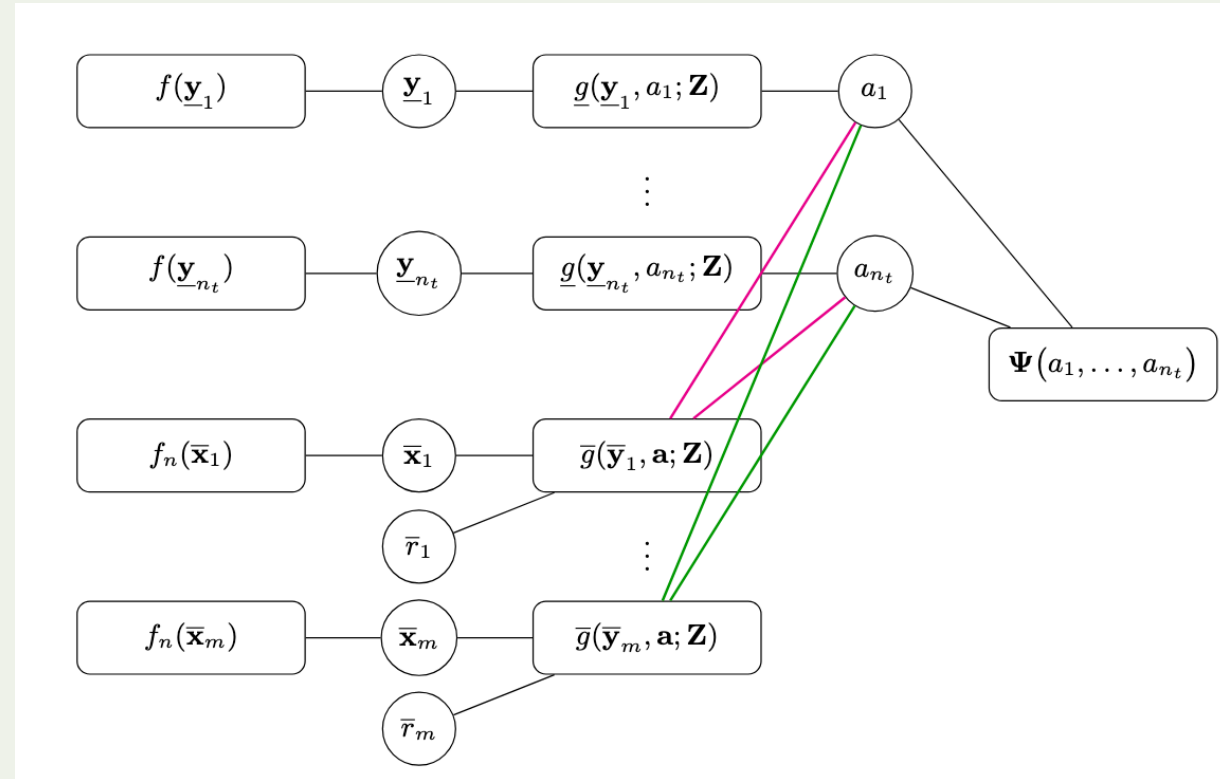
$$f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a} \mid Z) \propto \psi(\mathbf{a}) \prod_{i=1}^{n_t} f(\underline{\mathbf{y}}_i) \underline{\mathbf{g}}(\underline{\mathbf{y}}_i, a_i; Z) \\ \times \prod_{j=1}^m \bar{g}(\overline{\mathbf{y}}_j, \mathbf{a}; \mathbf{z}_j).$$

# New candidates share the association decisions.

Legacy chains remain on the left.

New candidates add  $\bar{g}$  factors.

The colored edges prevent a legacy claim and a new-object claim from explaining the same point.



Original Scenario 2 graph.

# Let every measurement name its source.

The extended state is

$$\mathbf{y} = (\mathbf{x}, \mathbf{e}).$$

Several measurements can share one object. Each measurement has exactly one source;  $b_i = 0$  means clutter.

$$b_i \in \{0, 1, \dots, K\}, \quad \mathcal{T}^k = \{i : b_i = k\}$$

$$p(|\mathcal{T}^k| \mid \mathbf{y}_k) = \frac{\mu_m(\mathbf{y}_k)^{|\mathcal{T}^k|}}{|\mathcal{T}^k|!} e^{-\mu_m(\mathbf{y}_k)}$$

# The allocation binomials telescope.

Each object selects its cell from the remaining measurements.

The allocation contributes a factorial for each cell, cancelling the Poisson factorials.

$$\prod_{k=1}^K \binom{n_c + \sum_{k'=k}^K |\mathcal{T}^{k'}|}{|\mathcal{T}^k|}^{-1} = \frac{n_c! \prod_{k=1}^K |\mathcal{T}^k|!}{m!},$$

$$\mu_c^{n_c} \prod_k \mu_m(\mathbf{y}_k)^{|\mathcal{T}^k|} = \mu_c^m \prod_k \left( \frac{\mu_m(\mathbf{y}_k)}{\mu_c} \right)^{|\mathcal{T}^k|}.$$

# One unary factor, then one edge per point

The zero-count exponential belongs to the object's unary factor.

An assigned point contributes a measurement-rate likelihood ratio. Every other edge is neutral.

$$g(\mathbf{y}_k) = f(\mathbf{y}_k)e^{-\mu_m(\mathbf{y}_k)}$$

$$g_{ki}(\mathbf{y}_k, b_i; \mathbf{z}_i) = \begin{cases} \frac{\mu_m(\mathbf{y}_k)f(\mathbf{z}_i | \mathbf{y}_k)}{\mu_c f_c(\mathbf{z}_i)}, & b_i = k, \\ 1, & b_i \neq k. \end{cases}$$

# Put all edges inside the object product.

Each object contributes its unary and all  $m$  pairwise factors.

All objects meet at the measurement-oriented association variables.

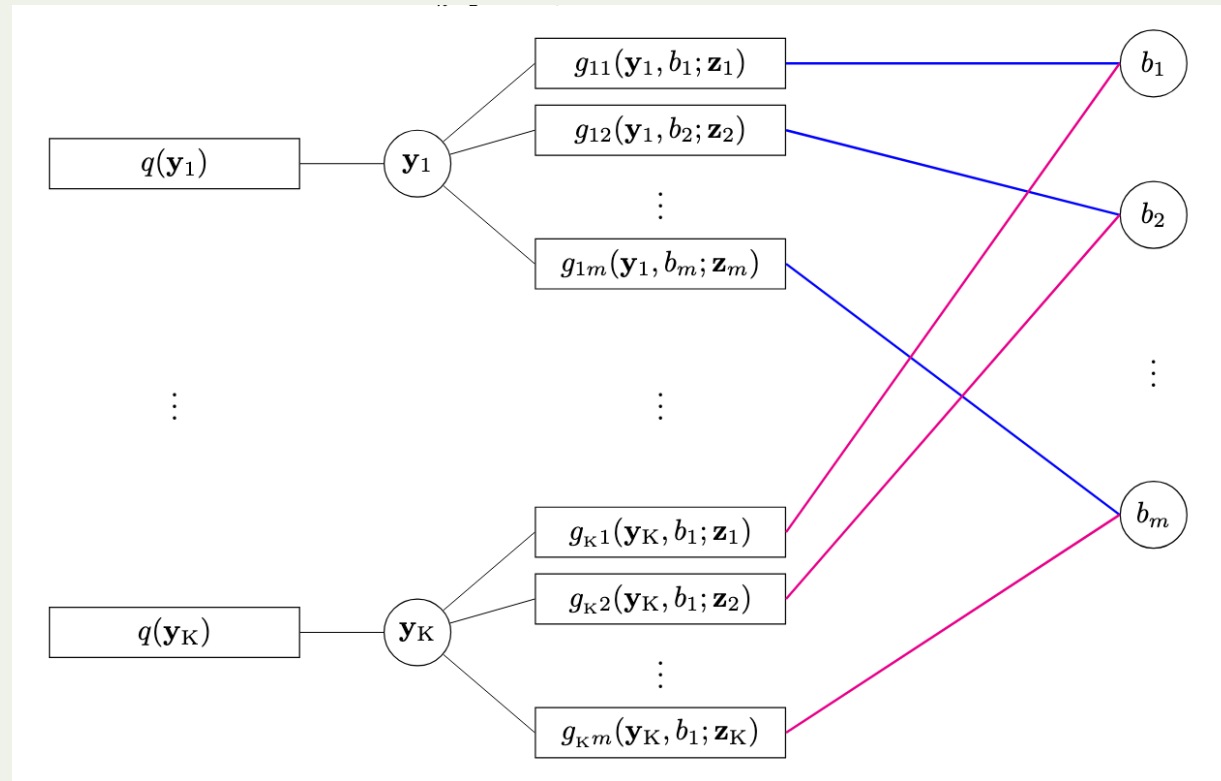
$$f(Y, \mathbf{b} \mid Z) \propto \prod_{k=1}^K \left[ g(\mathbf{y}_k) \prod_{i=1}^m g_{ki}(\mathbf{y}_k, b_i; \mathbf{z}_i) \right]$$

# Many measurements may connect to the same object.

Each object–measurement pair has a factor  $g_{ki}$ .

Variable  $b_i$  selects the source of measurement  $i$ .

The source choice itself gives each measurement one owner.



Original Scenario 3 graph.

# Anchor each new object to its first point.

$\underline{K}$  legacy potential objects and  
 $M$  new potential objects.

New candidate  $k$  is anchored at  
 measurement  $k$  and can explain  
 later measurements.

$$b_i \in \{0, 1, \dots, \underline{K} + i\},$$

$$\bar{r}_k = 1 \iff b_k = \underline{K} + k,$$

$$\bar{l}_k = \sum_{i=1}^M \mathbf{1}\{b_i = \underline{K} + k\}.$$

# A newly detected object has a nonzero count.

Newly detected objects use a zero-truncated Poisson count.

Existing legacy objects use an ordinary Poisson count; nonexistent ones generate zero points.

$$p(l \mid \mathbf{x}, \mathbf{e}, l \geq 1) = \frac{\mu_m(\mathbf{x}, \mathbf{e})^l}{l!} \frac{e^{-\mu_m(\mathbf{x}, \mathbf{e})}}{1 - e^{-\mu_m(\mathbf{x}, \mathbf{e})}}$$

$$p(l \mid \mathbf{y}) = \begin{cases} \frac{\mu_m(\mathbf{x}, \mathbf{e})^l}{l!} e^{-\mu_m(\mathbf{x}, \mathbf{e})}, & r = 1, \\ \mathbf{1}\{l = 0\}, & r = 0. \end{cases}$$

# Expand first, then remove the auxiliaries.

1. Introduce newly detected states and their associations.
2. Map them onto measurement-anchored candidates.
3. Integrate copied states; sum auxiliary labels and mappings.

$\tilde{\mathbf{y}}$  : newly detected object states,

$\tilde{\mathbf{b}}$  : their measurement assignments,

$\pi(k)$  : new PO  $k \mapsto$  newly detected object.

# Relabeling cancels the new-object factorial.

Permuting the auxiliary labels of  $\overline{K}$  newly detected objects gives the same canonical assignment. The sum contributes  $\overline{K}!$ .

$$\frac{\overline{K}! l_{\text{fa}}! \prod l_k!}{M!} \frac{\mu_n^{\overline{K}} e^{-\mu_n}}{\overline{K}!} \frac{\mu_{\text{fa}}^{l_{\text{fa}}} e^{-\mu_{\text{fa}}}}{l_{\text{fa}}!} \prod \frac{1}{l_k!} = \frac{\mu_n^{\overline{K}} \mu_{\text{fa}}^{l_{\text{fa}}} e^{-(\mu_n + \mu_{\text{fa}})}}{M!}$$

$$\overline{K} = |\mathcal{D}_b|, \quad l_{\text{fa}} = M - \sum_{k=1}^{\overline{K}} l_k - \sum_{k=1}^M \bar{l}_k$$

The count model is complete; values come next.

$\mathcal{D}_b$ : existing new candidates.  $\mathcal{D}_y$ : existing legacy objects.

$$\begin{aligned}
 f(\bar{\mathbf{y}}, \mathbf{b}, M \mid \underline{\mathbf{y}}) &= \frac{\mu_n^{|\mathcal{D}_b|} \mu_{\text{fa}}^{l_{\text{fa}}} e^{-(\mu_n + \mu_{\text{fa}})}}{M!} \\
 &\times \prod_{k \in \mathcal{D}_b} \left[ \bar{r}_k f_n(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k) \mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)^{\bar{l}_k} \frac{e^{-\mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)}}{1 - e^{-\mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)}} \right] \\
 &\times \prod_{k \notin \mathcal{D}_b} \left[ (1 - \bar{r}_k) f_d(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k) \mathbf{1}\{\bar{l}_k = 0\} \right] \\
 &\times \prod_{k \in \mathcal{D}_y} \left[ \mu_m(\underline{\mathbf{x}}_k, \underline{\mathbf{e}}_k)^{l_k} e^{-\mu_m(\underline{\mathbf{x}}_k, \underline{\mathbf{e}}_k)} \right] \prod_{k \notin \mathcal{D}_y} \mathbf{1}\{l_k = 0\}.
 \end{aligned}$$

# Separate legacy prediction and new-object evidence.

$f^-(\underline{\mathbf{y}})$  includes the legacy existence mass and its conditional state density.

The new-object unary carries the detected-object intensity and the truncated-count normalization.

$$\underline{q}(\underline{\mathbf{y}}) = \begin{cases} f^-(\underline{\mathbf{y}})e^{-\mu_m(\underline{\mathbf{x}}, \underline{\mathbf{e}})}, & \underline{\mathbf{r}} = 1, \\ f^-(\underline{\mathbf{y}}), & \underline{\mathbf{r}} = 0. \end{cases}$$

$$\bar{q}(\bar{\mathbf{y}}) = \begin{cases} \mu_n f_n(\bar{\mathbf{x}}, \bar{\mathbf{e}}) \frac{e^{-\mu_m(\bar{\mathbf{x}}, \bar{\mathbf{e}})}}{1 - e^{-\mu_m(\bar{\mathbf{x}}, \bar{\mathbf{e}})}}, & \bar{\mathbf{r}} = 1, \\ f_d(\bar{\mathbf{x}}, \bar{\mathbf{e}}), & \bar{\mathbf{r}} = 0. \end{cases}$$

# An absent object cannot own a point.

Write the rate-weighted likelihood ratio as  $\rho_{ki}$ .

Assignment to an existing object activates that ratio; an absent object forbids the assignment.

$$\rho_{ki} = \frac{\mu_m(\mathbf{x}_k, \mathbf{e}_k) f(\mathbf{z}_i \mid \mathbf{x}_k, \mathbf{e}_k)}{\mu_{\text{fa}} f_{\text{fa}}(\mathbf{z}_i)}$$

$$\underline{g}_{ki}(\underline{\mathbf{y}}_k, b_i; \mathbf{z}_i) = \begin{cases} \rho_{ki}, & b_i = k, \underline{\mathbf{r}}_k = 1, \\ 1, & b_i \neq k, \underline{\mathbf{r}}_k = 1, \\ \mathbf{1}\{b_i \neq k\}, & \underline{\mathbf{r}}_k = 0. \end{cases}$$

# The anchor enforces existence in both directions.

If candidate  $k$  exists,  
measurement  $k$  must belong to it.

If it does not exist, the anchor  
cannot name it.

$$\bar{g}_k(\bar{\mathbf{y}}_k, b_k; \mathbf{z}_k) = \begin{cases} \rho_{kk}, & b_k = \underline{K} + k, \bar{r}_k = 1, \\ 0, & b_k \neq \underline{K} + k, \bar{r}_k = 1, \\ \mathbf{1}\{b_k \neq \underline{K} + k\}, & \bar{r}_k = 0. \end{cases}$$

# An extended new object needs more than its anchor.

For every  $i > k$ , candidate  $k$  can explain measurement  $i$  through  $\bar{h}_{ki}$ .

Earlier measurements cannot refer to a later candidate.

$$\bar{h}_{ki}(\bar{\mathbf{y}}_k, b_i; \mathbf{z}_i) = \begin{cases} \rho_{ki}, & b_i = \underline{K} + k, \bar{r}_k = 1, \\ 1, & b_i \neq \underline{K} + k, \bar{r}_k = 1, \quad i > k \\ \mathbf{1}\{b_i \neq \underline{K} + k\}, & \bar{r}_k = 0. \end{cases}$$

# Two families of local products

Legacy objects connect to every measurement. New candidates connect to their anchor and all later measurements.

$$f(\underline{\mathbf{y}}, \bar{\mathbf{y}}, \mathbf{b} \mid \mathbf{Z}) \propto \prod_{k=1}^K \left[ \underline{q}(\underline{\mathbf{y}}_k) \prod_{i=1}^M \underline{g}_{ki}(\underline{\mathbf{y}}_k, b_i; \mathbf{z}_i) \right]$$

$$\times \prod_{k=1}^M \left[ \bar{q}(\bar{\mathbf{y}}_k) \bar{g}_k(\bar{\mathbf{y}}_k, b_k; \mathbf{z}_k) \prod_{i=k+1}^M \bar{h}_{ki}(\bar{\mathbf{y}}_k, b_i; \mathbf{z}_i) \right].$$

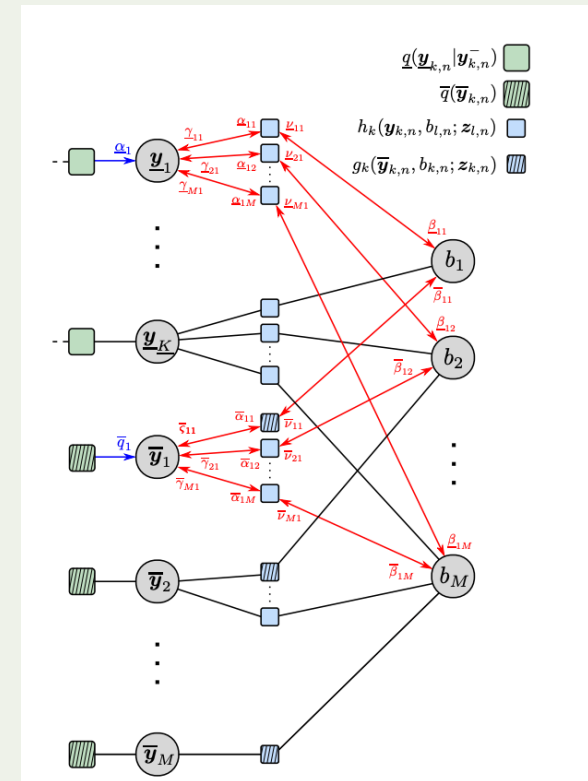
# Prediction, birth, and measurement edges

$\underline{q}$ : legacy prediction.

$\bar{q}$ : new-object unary.

The anchor and later edges encode the canonical labeling.

Red annotations identify BP messages on this graph.



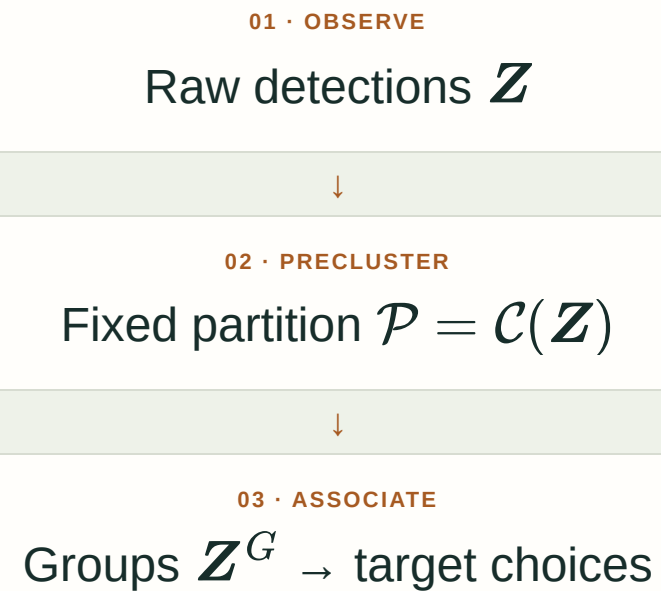
Original Scenario 4 graph. Its  $h / \bar{g}$  labels follow the source figure.

# Derivation for GrBP

GrBP uses grouped measurements: external clustering fixes one partition.

Data association then decides which target explains each **whole group**.

Introduce one newborn candidate per group.



# Points, groups, and targets are different counts.

$m$  raw detections form  $n^g$  disjoint groups.

$n^t$  legacy components carry predicted beliefs.

Every component has kinematics, extent, and existence;  
 $s = (\underline{x}, \underline{E})$  is shorthand.

$$\mathbf{Z}^G = \{\mathbf{Z}^1, \dots, \mathbf{Z}^{n^g}\}, \quad m = \sum_{j=1}^{n^g} |\mathbf{Z}^j|.$$

$$s = (\underline{x}, \underline{E}), \quad \underline{\mathbf{y}}^p = (\underline{s}^p, \underline{r}^p),$$

$$f(\underline{\mathbf{Y}}) = \prod_{p=1}^{n^t} f(\underline{\mathbf{y}}^p).$$

# Two variables record the same claim.

$a^p = j$ : legacy target  $p$  claims group  $j$ .

$b^j = p$ : group  $j$  comes from legacy target  $p$ .

$b^j = 0$  leaves two alternatives:  
**newborn or clutter.**

$$a^p \in \{0, \dots, n^g\}, \quad b^j \in \{0, \dots, n^t\},$$

$$a^p = j \iff b^j = p.$$

$$\psi(\mathbf{a}, \mathbf{b}) = \prod_{p=1}^{n^t} \prod_{j=1}^{n^g} \Psi_{p,j}(a^p, b^j),$$

$$\Psi_{p,j}(a, b) = \mathbf{1}[(a = j) \iff (b = p)].$$

# Read the counts from the assignments.

Detected legacy targets, existing newborns, and clutter partition the groups on every feasible hypothesis.

$$\begin{aligned}\mathbb{D} &= \{p : a^p \neq 0\}, & \mathbb{N} &= \{j : \bar{r}^j = 1\}, \\ \mathbb{C} &= \{1, \dots, n^g\} \setminus (\{a^p : p \in \mathbb{D}\} \cup \mathbb{N}), \\ n^d &= |\mathbb{D}|, & n^n &= |\mathbb{N}|, & n^c &= n^g - n^d - n^n.\end{aligned}$$

# Separate the prior, allocation, and likelihood.

Condition throughout on the fixed partition  $\mathcal{P}$ . Its notation is suppressed in the source.

$$\begin{aligned} f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}, \mathbf{Z}^G) &= f(\underline{\mathbf{Y}}) \psi(\mathbf{a}, \mathbf{b}) \\ &\quad \times f(\mathbf{a}, \mathbf{b}, \overline{\mathbf{Y}} \mid \underline{\mathbf{Y}}) \\ &\quad \times f(\mathbf{Z}^G \mid \underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}). \end{aligned}$$

# Clutter groups have their own Poisson mean.

$\mu_g$ : mean number of clutter **groups**.

$\mu_n$ : mean number of **detected newborns**.

Generally  $\mu_g \neq \mu_c$ : a clutter group can contain several detections.

$$f(n^c) \approx \frac{\mu_g^{n^c} e^{-\mu_g}}{n^c!},$$

$$f(n^n) = \frac{\mu_n^{n^n} e^{-\mu_n}}{n^n!}.$$

# The allocation factorials cancel the count factorials.

Choose which unclaimed groups are newborns.

Assign detected legacy targets to distinct group slots.

Multiply by the two Poisson count laws; the remaining denominator is  $n^g!$ .

$$\begin{aligned} \binom{n^c + n^n}{n^n}^{-1} &= \frac{n^c! n^n!}{(n^c + n^n)!}, \\ \frac{1}{n^g P_{n^d}} &= \frac{(n^c + n^n)!}{n^g!}, \\ \text{product} &= \frac{n^c! n^n!}{n^g!}. \end{aligned}$$

# Keep the existence branches and newborn densities.

$B$  collects newborn state densities;  $f_d$  is normalized. The nonempty legacy count mass enters the likelihood next.

$$f(\mathbf{a}, \mathbf{b}, \overline{\mathbf{Y}} \mid \underline{\mathbf{Y}}) \approx \frac{\mu_g^{n^c} e^{-\mu_g} \mu_n^{n^n} e^{-\mu_n}}{n^g!} B(\overline{\mathcal{X}} \mid \overline{\mathbf{r}}) \\ \times \prod_{p \in \mathbb{D}} \underline{r}^p \\ \times \prod_{p \notin \mathbb{D}} \left[ 1 - \underline{r}^p \left( 1 - e^{-\mu_m(\underline{s}^p)} \right) \right].$$

$$B(\overline{\mathcal{X}} \mid \overline{\mathbf{r}}) = \prod_{j \in \mathbb{N}} f_n(\overline{s}^j) \prod_{j \notin \mathbb{N}} f_d(\overline{s}^j)$$

# A detected newborn uses a nonempty-group likelihood.

A legacy group uses the extended-target Poisson likelihood.

A newborn is detected by definition, so condition on a nonzero count.

Keep one exponential per target, not one per point.

$$L_L(\mathbf{Z}^j \mid s) = e^{-\mu_m(s)} \prod_{\mathbf{z} \in \mathbf{Z}^j} \mu_m(s) f(\mathbf{z} \mid s).$$

$$L_N(\mathbf{Z}^j \mid s) = \frac{L_L(\mathbf{Z}^j \mid s)}{1 - e^{-\mu_m(s)}}, \quad \mathbf{Z}^j \neq \emptyset.$$

# Multiply the group likelihoods by source.

Conditioned on a feasible association, the approximate model treats groups as independent.

$$f(\mathbf{Z}^G \mid \underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}) = \prod_{p \in \mathbb{D}} L_{\mathbf{L}}(\mathbf{Z}^{a^p} \mid \underline{\mathbf{s}}^p) \\ \times \prod_{j \in \mathbb{N}} L_{\mathbf{N}}(\mathbf{Z}^j \mid \overline{\mathbf{s}}^j) \prod_{c \in \mathbb{C}} F_c(\mathbf{Z}^c).$$

# Divide each target group by its clutter explanation.

The product of all group clutter densities is a baseline for the entire scan.

Redistributing  $\mu_g^{n^c}$  leaves **one**  $1/\mu_g$  for every target-generated group.

$$F_c(\mathbf{Z}^j) = \prod_{\mathbf{z} \in \mathbf{Z}^j} f_c(\mathbf{z}),$$

$$\prod_{j=1}^{n^g} F_c(\mathbf{Z}^j) = \prod_{i=1}^m f_c(\mathbf{z}^i).$$

$$\mu_g^{n^c} = \mu_g^{n^g} \prod_{p \in \mathbb{D}} \mu_g^{-1} \prod_{j \in \mathbb{N}} \mu_g^{-1}.$$

# Separate the constant and define the group ratio.

$C$  is fixed for the observed grouped scan and fixed model parameters.

$R_j(s)$  abbreviates the per-point likelihood ratios inside group  $j$ .

$$C(\mu_g, \mu_n, \mathbf{Z}^G) = \frac{\mu_g^{n^g} e^{-\mu_g} e^{-\mu_n}}{n^g!} \prod_{i=1}^m f_c(\mathbf{z}^i).$$

$$R_j(s) = \prod_{\mathbf{z} \in \mathbf{Z}^j} \frac{\mu_m(s) f(\mathbf{z} \mid s)}{f_c(\mathbf{z})}.$$

# A legacy target has four local cases.

An existing target can claim a group or generate no detections.

A nonexistent target can only make a null claim.

The predicted prior remains a separate factor.

$$\underline{l}((s, r), a; \mathbf{Z}^G) = \begin{cases} \frac{e^{-\mu_m(s)}}{\mu_g} R_a(s), & a \neq 0, r = 1, \\ e^{-\mu_m(s)}, & a = 0, r = 1, \\ 1, & a = 0, r = 0, \\ 0, & a \neq 0, r = 0. \end{cases}$$

# An unclaimed group can create one newborn.

$b^j = 0$ : newborn and clutter compete.

$b^j \neq 0$ : a legacy target already owns the group, so the newborn cannot exist.

The absent candidate carries dummy density  $f_d$ .

$$\bar{l}((s, r), b; \mathbf{Z}^j) = \begin{cases} \frac{\mu_n f_n(s) e^{-\mu_m(s)}}{\mu_g [1 - e^{-\mu_m(s)}]} R_j(s), & b = 0, r = 1, \\ 0, & b \neq 0, r = 1, \\ f_d(s), & r = 0. \end{cases}$$

# The grouped joint PDF becomes a product of local factors.

Predicted priors  $\times$  legacy factors  $\times$  newborn factors  $\times$  group-association consistency.

$$\begin{aligned}
 f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}, \mathbf{Z}^G) &\approx C \psi(\mathbf{a}, \mathbf{b}) \\
 &\times \prod_{p=1}^{n^t} \left[ f(\underline{\mathbf{y}}^p) \underline{l}(\underline{\mathbf{y}}^p, \mathbf{a}^p; \mathbf{Z}^G) \right] \\
 &\times \prod_{j=1}^{n^g} \overline{l}(\overline{\mathbf{y}}^j, \mathbf{b}^j; \mathbf{Z}^j).
 \end{aligned}$$

# Association cannot repair a fixed partition.

**S4:** infer assignments of individual detections; several points can choose the same target.

**GrBP:** infer whole-group assignments after external clustering.

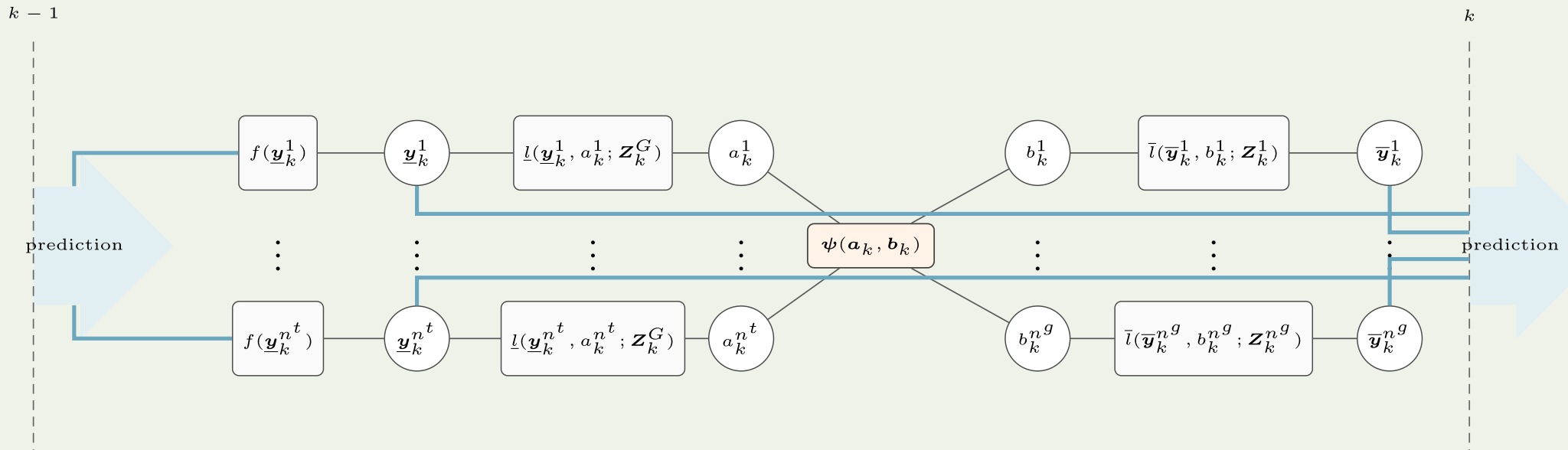
**One group per target.  
One source per group.**

Target splits and cross-target merges are outside this model.

The number of groups is not the inferred number of targets.

# Factor graph for GrBP

Legacy chains and group-indexed newborn chains meet at  $\psi(\mathbf{a}_k, \mathbf{b}_k)$ .



# From raw measurements to fixed groups

**Unknown object count:**

introduce potential objects and existence.

**Multiple points per object:** use measurement-oriented assignments and Poisson counts.

**GrBP:** precluster once, then associate whole groups.

| Case | Association          | Coupling                       |
|------|----------------------|--------------------------------|
| S1   | Target-oriented      | Global exclusivity             |
| S2   | Target-oriented      | Exclusivity + new objects      |
| S3   | Measurement-oriented | Object–point factors           |
| S4   | Measurement-oriented | Legacy + anchor + later edges  |
| GrBP | Group-level          | Exclusive target–group factors |

# Check the tiny cases first.

1. Keep ordered-vector and finite-set conventions consistent.
2. Retain every state-dependent factor.
3. Verify counts and normalization by enumeration.

1. Keep  $\mu_n f_n$  in new-object factors.
2. Enforce the anchor condition and include every later edge.
3. Compare approximate BP marginals against exact inference on small scans.

# The page, the scans, and the primary papers

## R1 · Point-object models

Meyer et al., “Message passing algorithms for scalable multitarget tracking,” Proceedings of the IEEE, 2018.

## R2 · Extended-object model

Granström, Baum & Reuter, “Extended Object Tracking: Introduction, Overview and Applications.”

## R3 · Extended objects with uncertain count

Meyer & Williams, “Scalable Detection and Tracking of Geometric Extended Objects,” 2021.

## R4 · Preclustering and group association

EO\_Writing\_Long\_Version, Appendix I, Eqs. (A.1)–(A.9).

The companion page contains the full derivations and original handwritten scans for S1–S4, plus the Appendix I derivation for GrBP.